

Milestone 2 Technical Evaluation Report

Competition Link:Smart MCQ Solver Challenge

Account Identity:23f2005639@ds.study.iitm.ac.in

Core Target:Deep Learning Foundations, Attention Mechanics, Context Embeddings, & Zero-Shot Verification

Objective Overview: This report marks the transition from token-frequency mapping methods to dense multi-layered context configurations. It addresses the validation metrics, token parameters, hidden state measurements, and zero-shot engineering performance evaluated across the 2,000 observations in the competition dataset.

1. Hugging Face Datasets & Tokenizer Mechanics

Data integration pipelines were refactored using the Hugging Face ecosystem to enable unified token mapping routines instead of manual string parsing techniques.

Evaluated Tokenizer / Dataset Metric	Verified Technical Value
String character length of <code>combined_text</code> (Index 51)	614 characters
Hardcoded subword vocabulary dictionary size (<code>bert-base-uncased</code>)	30,522 unique tokens
Integer vocabulary map alignment index for <code>[SEP]</code>	102
Geometric dimension footprint of padded <code>input_ids</code> tensor	[2000, 128] (Batch, Sequence Length)

2. Transformer Architecture & Attention State Audits

The inner tensor states of a standard multi-head attention block were mapped and calculated using a baseline `bert-base-uncased` instantiation. The hidden state dimension (\$768\$) is distributed uniformly across \$12\$ parallel attention pathways.

- Attention Head Dimensionality:** Each separate tracking head handles an internal layout size of exactly $768 / 12 = 64$ hidden elements.
- Last Hidden State Vector Space:** Tokenizing row ID 0 under default constraints and passing it through the layer stack generates an exact tensor footprint of `[1, 31, 768]`.
- Classification Vector Matrix Sum:** Accumulating the initial 5 float metrics inside the row 0 classification token vector (`[CLS]` at index 0) yields a sum of **-1.2001**.

- **Self-Attention Token Routing:** For the evaluation sequence *"Light-ion fusion is a technique."*, the precise self-attention weight running from the [CLS] token to the keyword entry *"fusion"* in the final layer's first head is **0.1025**.

3. Dense Context Embeddings & Semantic Search Validation

To prevent the lexical matching dropouts identified in Milestone 1, context-aware embeddings were generated using `sentence-transformers/all-MiniLM-L6-v2`. Calculating vector mappings for row ID 0 yields a high prompt-to-option similarity metric of **0.7658**.

all-MiniLM-L6-v2 Baseline MAP@3 Performance

0.423083

Incremental Correct Predictions Captured

564 Entries

Error Vector Recovery: Running a comparative delta analysis confirms that the dense context pipeline correctly retrieved **564 questions** within its Top-3 ranking window that were completely missed or discarded by the statistical TF-IDF string matching strategy.

4. Zero-Shot Logit Space & Generative SLM Verification

- **Natural Language Inference Probabilities:** Processing row index 1 through a zero-shot pipeline (`facebook/bart-large-mnli`) evaluates option validities to return a top-ranked candidate score of **0.4575**.
- **Multi-Label Sigmoid Delta:** Disabling standard Softmax normalization and switching layers to independent unconstrained sigmoids (`multi_label=True`) results in a total probability summation delta of exactly **0.9995**.
- **Small Language Model Prompt Routing:** Generative question answering formatting was validated using `google/flan-t5-small`. The architecture successfully routed prompt tokens through its generative text-to-text loop.

Verified SLM Prompt/Generation Sequence Trace:

Question: Pick the best possible answer: What is Martin Heidegger's view on the relationship between time and human existence? among the listed options.. Is the correct answer A: Martin Heidegger believes that humans exist within a time continuum that is infinite and does not have a defined beginning or end. The relationship to the past involves acknowledging it as a historical era, and the relationship to the future involves creating a world that will endure beyond one's own time. or B: Martin Heidegger believes that humans do not exist inside time, but that they are time. The relationship to the past is a present awareness of having been, and the relationship to the future involves anticipating a potential possibility, task, or engagement.? Answer with just the letter A or B.